MIL-C-12038C(EA)
2 June 1982
SUPERSEDING
MIL-C-12038B
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MILITARY SPECIFICATION

CHLOROBENZENE, TECHNICAL

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one grade of monochlorobenzene (C_6H_5Cl).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-D-729 - Drums, Shipping and Storage, Steel, 55-Gallon (208 Liters)
PPP-P-704 - Pails, Metal: (Shipping, Steel, 1 Through 12 Gallons)

FSC 6810

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

2.1.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 171 to 179 - Department of Transportation Hazardous Materials Regulations

(The Code of Federal Regulations is available from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. Orders for the above publication should cite "49 CFR 171 to 179".)

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DOD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

UNIFORM FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

ASTM STANDARDS

D86 - Distillation of Petroleum Products
D941 - Density and Relative Density (Specific Gravity) of Liquids by Lipkin Bicapillary Pycnometer
D1193 - Reagent Water
D1744 - Water in Liquid Petroleum Products by Karl Fischer Reagent

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

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(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Color. The chlorobenzene shall be clear and colorless when tested as specified in 4.2.4.1.

3.2 Chemical and physical characteristics. The chlorobenzene shall conform to the chemical and physical characteristics of table I when tested as specified therein.

TABLE I. Chemical and physical characteristics

Characteristic	Minimum	Maximum	Test paragraph
Specific gravity at 20°/4°C	1.101	1.112	4.2.4.2
Acidity, as HCL, percent by weight		0.015	4.2.4.3
Water, percent by volume		0.1	4.2.4.4
Distillation range (131° to 132°C at 760 millimeters Hg, percent by volume distilled away)	98		4.2.4.5

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the chlorobenzene produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

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4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For chlorobenzene test. Sampling shall be conducted in accordance with table II. A representative specimen of approximately 300 milliliters (ml) shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken. Do not use any plastic container or plastic lids; chlorobenzene will break these down very rapidly.

TABLE II. Sampling for chlorobenzene test

Number of containers in batch or lot	Number of sample containers
2 to 25	2
26 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

4.2.2.3 For container leakage test. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one filled unit, intermediate, or shipping container, as applicable, ready for shipment. Sample unit, intermediate, and shipping containers shall be examined for the following defects using an AQL 2.5 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged or leaking
(e) Marking incorrect, missing, or illegible
(f) Gaskets missing or not as specified
(g) Dust covers missing or not as specified

4.2.3.2 For chlorobenzene test. Approximately equal portions of all of the specimens taken in 4.2.2.2 shall be thoroughly mixed to form a composited specimen of no less than 600 ml. The composited specimen shall be tested as specified in 4.2.4. Each test shall be conducted in duplicate analysis. Failure of either analysis of any test shall be cause for rejection of the lot represented.

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4.2.3.3 For container leakage test. The sample unit shall be one container. The sample containers selected in 4.2.2.3 shall be tested as specified in 4.2.5 using an AQL of 1.5 percent defective.

4.2.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Color. Visually examine the chlorobenzene to determine clarity and color.

4.2.4.2 Specific gravity. Determine the specific gravity of the specimen in accordance with ASTM D941, at a test temperature of 20°C and a reference temperature of 4°C.

4.2.4.3 Acidity. Measure accurately 25 ml of the specimen into a 250-ml glass-stoppered Erlenmeyer flask. Add 25 ml of neutralized water, and 1 ml of phenolphthalein indicator to the specimen. Stopper the flask and shake for 30 seconds. Titrate the contents of the flask with 0.02 N sodium hydroxide. The pink color, upon shaking, shall remain in the aqueous layer for 1 minute before the end point is reached. Calculate the percent acidity as follows:

$$\text{Percent acidity, as HCl} = \frac{36.5\ AB}{CD}$$

where: A = Milliliters of sodium hydroxide used
B = Normality of sodium hydroxide solution
C = Milliliters of specimen used
D = Specific gravity of specimen (from 4.2.4.2).

4.2.4.4 Water content. Determine the water content of the specimen in accordance with ASTM D1744.

4.2.4.5 Distillation range. Determine the distillation range of the specimen in accordance with ASTM D86.

4.2.5 Container leakage test. Place the container in each of the following positions, and leave it in each for a period of 15 minutes.

(a) Upright
(b) Upside down
(c) On one side
(d) On other side

Examine the container after each period for any evidence of leakage.

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5. PACKAGING

5.1 Unit packing. Chlorobenzene shall be unit packed level A, B or C as specified (see 6.2).

5.1.1 Level A. Chlorobenzene shall be furnished in a 45 or 500-pound quantity as specified (see 6.2).

5.1.1.1 Forty-five pound quantity. A quantity of 45 (+0.5 or -0) pounds of chlorobenzene shall be unit packed level A in a clean, dry nominal 5-gal capacity pail conforming to type I, class 4 of PPP-P-704. The pail shall be furnished with a push-pull spout and closure seal, as well as a metal screwcap. The closure seal and screwcap shall be applied to the pail nozzle in accordance with the pail manufacturer's instructions. The pail shall show no evidence of leakage when tested as specified in 4.2.5.

5.1.1.2 Five-hundred pound quantity. A quantity of 500 (+5 or -0) pounds of chlorobenzene shall be unit packed in accordance with Department of Transportation (DOT) regulations, in a nominal 55-gal capacity steel drum. The drum shall conform to DOT specification 17C. The entire inner and exterior surface of the drum shall be cleaned and phosphatized and that of the exterior painted as specified in PPP-D-729. The interior surface shall be fog sprayed with rust preventative at the option of the procuring activity as specified in PPP-D-729 (see 6.2). The plug closures shall be furnished with gaskets, serviceable in preventing leakage of chlorobenzene. The drum shall be filled only after it has been determined by careful inspection to be free of moisture, rust and other contamination. The drum shall be securely closed immediately after filling by threaded plugs torqued tight to within a range as specified by the drum manufacturer. Each drum shall be fitted with pilfer-proof seals over each plug and flange.

5.1.2 Level B. The chlorobenzene shall be unit packed level B as specified for level A except that standard commercial painting of the pail and drum exterior instead of that specified by PPP-D-704 and PPP-D-729 shall be acceptable, and the pail as a minimum shall conform to type I, class 3 of PPP-P-704, and the drum as a minimum shall conform to type II of PPP-D-729.

5.1.3 Level C. The chlorobenzene shall be unit packed level C in the quantity as specified for level A, in a clean, dry drum to assure protection against deterioration of contents and drum from supply source to the first receiving activity and for a minimum period of six months. The drum shall comply with DOT regulations, Uniform Freight Classification Rules and National Motor Freight Classification Rules, or those of other common carriers applicable to the mode of transportation.

5.2 Packing. Chlorobenzene unit packed as specified in 5.1 shall require no further protection for shipment.

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5.3 Marking. Containers shall be marked in accordance with MIL-STD-129, DOT regulations and any other applicable regulations. In addition, unit containers shall be marked with the following precautionary marking:

HAZARD: Flammable Liquid

WARNING! FLAMMABLE!

Keep away from heat, sparks, and open flames.
Keep container closed when not in use.
Vapor may explode if ignited in an enclosed area.
Extinguish with dry chemical, foam, or carbon dioxide. Do not use water.
For spills, ventilate, extinguish flames, and soak up with sand, earth, or cat litter.
Use only with adequate ventilation and respirator, gloves, and face shield. If used in confined space, wear self-contained breathing apparatus.
Avoid contact with liquid and vapor.
Vapor not irritating to eyes, nose, or throat, but is harmful.
Liquid irritating to skin and eyes.
Flush affected areas with plenty of water.
If swallowed, dilute by drinking water. Call a physician.

Special Storage Instructions:

Protect against physical damage.
Inside storage should be in a standard fire-resistive flammable liquids storage room or cabinet.
SEPARATE FROM OXIDIZING MATERIALS.

6. NOTES

6.1 Intended use. The chlorobenzene covered by this specification is intended for use as a solvent and as a raw material in the manufacture of various chemicals.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification
(b) Level of unit packing required (see 5.1)
(c) Quantity of chlorobenzene required (see 5.1.1)
(d) If fog spray with rust preventative is required (see 5.1.1.2)

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

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6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E29.

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Preparing activity:

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